Лабораторная работа №3

Моделирование сетей передачи данных

Исаев Б.А.

2024

Российский университет дружбы народов имени Патриса Лумумбы, Москва, Россия

Докладчик

- Исаев Булат Абубакарович
- Студент группы НПИбд-01-22
- Студ. билет 1132227131
- Российский университет дружбы народов имени Патриса Лумумбы

Цель лабораторной работы

- Познакомиться с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получить навыки проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

Создание простейшей топологии сети

mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo

```
* Management:     https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Oct 31 11:08:14 2025
mininet@mininet-vm:~$ cd ~/work/lab_iperf3/
mininet@mininet-vm:~/work/lab_iperf3$ mkdir lab_iperf3_topo
mkdir: cannot create directory 'lab_iperf3_topo': File exists
mininet@mininet-vm:~/work/lab_iperf3$ ls
iperf3  iperf.csv  iperf_results.json  lab_iperf3_topo  results
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp ~/mininet/examples/emptynet.py
cp: missing destination file operand after '/home/mininet/mininet/examples/emptynet.py'
Try 'cp --help' for more information.
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp ~/mininet/examples/emptynet.py ~/work/lab_iperf3/lab_iperf3_topo/
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv emptynet.py lab_iperf3_topo
bmv: command not found
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv emptynet.py lab_iperf3_topo

Command 'Bmv' not found, did you mean:

  command 'imv' from deb renameutils (0.12.0-7)
  command 'qmv' from deb renameutils (0.12.0-7)
  command 'mmv' from deb mmv (1.01b-19build1)
  command 'mv' from deb coreutils (8.30-3ubuntu2)

Try: sudo apt install <deb name>
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv emptynet.py lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv emptynet.py lab_iperf3_topo.py
mv: cannot stat 'emptynet.py': No such file or directory
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv lab_iperf3_topo lab_iperf3_topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ touch lab_iperf3_topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ nano lab_iperf3_topo.py
```

Создание простейшей топологии сети

mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo

GNU nano 4.8 lab_iperf3_topo.py

```python
#!/usr/bin/env python

"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( '*** Starting network\n')
    net.start()

    info( '*** Running CLI\n' )
    CLI( net )

    info( '*** Stopping network' )
    net.stop()


if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
```

Figure 2: Открытие файла lab_iperf3_topo.py

Создание простейшей топологии сети

```
mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo
KeyboardInterrupt

mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
*** Running CLI
*** Starting CLI:
mininet> net
h1 h1-eth0:s3-eth1
h2 h2-eth0:s3-eth2
s3 lo:  s3-eth1:h1-eth0 s3-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s3-eth1 (OK OK)
h2-eth0<->s3-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=1308>
<Host h2: h2-eth0:10.0.0.2 pid=1310>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=1315>
<Controller c0: 127.0.0.1:6653 pid=1301>
mininet> exit
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$
```

Figure 3: Запуск скрипта создания топологии и дальнейший просмотр элементов

Внесение изменений в скрипт

Выбрать mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo

```
  GNU nano 4.8                    lab_iperf3_topo.py                    Modified
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( '*** Starting network\n')
    net.start()

    print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )

    info( '*** Running CLI\n' )
    CLI( net )

    info( '*** Stopping network' )
    net.stop()


if __name__ == '__main__':
    setLogLevel( 'info' )

^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos
^X Exit        ^R Read File   ^\ Replace     ^U Paste Text  ^T To Spell    ^_ Go To Line
```

Figure 4: Внесение изменения в скрипт, позволяющего вывести на экран информацию о хосте h1 (имя, IP-адрес, MAC-адрес)

Проверка

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller

*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address fe:1f:f1:a4:f7:1e
*** Running CLI
*** Starting CLI:
mininet>
mininet>
Interrupt
mininet> exit
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
```

Figure 5: Проверка корректности отработки скрипта

Внесение изменений в скрипт

mininet@mininet-vm: ~/work/lab_iperf3/lab_iperf3_topo

```
  GNU nano 4.8                    lab_iperf3_topo.py                    Modified
#!/usr/bin/env python

"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( '*** Starting network\n')
    net.start()

    print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )
    print( "Host", h2.name, "has IP address", h2.IP(), "and MAC address", h2.MAC() )

    info( '*** Running CLI\n' )
    CLI( net )

    info( '*** Stopping network' )
    net.stop()


if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()

^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Paste Text ^T To Spell   ^_ Go To Line M-E Redo
```

Figure 6: Внесение изменения в скрипт, позволяющего вывести на экран информацию о двух хостах (имя, IP-адрес, MAC-адрес)

Проверка

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ nano lab_iperf3_topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 96:37:e3:09:f6:ee
Host h2 has IP address 10.0.0.2 and MAC address 12:ce:b1:51:e2:cb
*** Running CLI
*** Starting CLI:
mininet>
```

Figure 7: Проверка корректности отработки скрипта

Добавление в скрипт настроек параметров производительности

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp lab_iperf3_topo.py lab_iperf3_topo2.py
```

Figure 8: Создание копии скрипта lab_iperf3_topo.py

Добавление в скрипт настроек параметров производительности

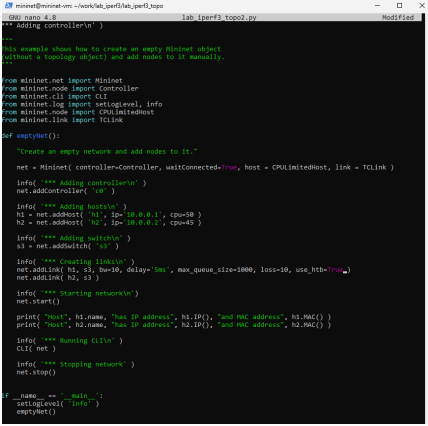

Figure 9: Изменение скрипта lab_iperf3_topo2.py: добавление ипорта классов, изменение строки описания сети, изменение функции задания параметров виртуального хоста h1 и h2, изменение функции параметров соединения между хостом h1 и коммутатором s3

Добавление в скрипт настроек параметров производительности

```
^[[Amininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ nano lab_iperf3_topo2.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$

mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo2.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(10.00Mbit 5ms delay 10.00000% loss) (10.00Mbit 5ms delay 10.00000% loss) *** Starting network
*** Configuring hosts
h1 (cfs 5000000/100000us) h2 (cfs 4500000/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (10.00Mbit 5ms delay 10.00000% loss) ...(10.00Mbit 5ms delay 10.00000% loss)
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 5a:21:3d:31:68:19
Host h2 has IP address 10.0.0.2 and MAC address ca:e1:40:91:6f:bd
*** Running CLI
*** Starting CLI:
mininet>
```

Figure 10: Запуск скрипта lab_iperf3_topo2.py на отработку

Добавление в скрипт настроек параметров производительности

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp lab_iperf3_topo2.py lab_iperf3.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mkdir -p ~/work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv ~/work/lab_iperf3/lab_iperf3_topo/lab_iperf3.py ~/
work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cd ~/work/lab_iperf3/iperf3/
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ ls -l
total 24
-rw-rw-r-- 1 mininet mininet  976 Oct  9 08:52 iperf.csv
-rw-r--r-- 1 mininet mininet 7779 Oct  9 08:52 iperf_result.json
-rwxrwxr-x 1 mininet mininet 1346 Oct 31 09:14 lab_iperf3.py
-rw-rw-r-- 1 mininet mininet  179 Oct  9 08:51 Makefile
drwxrwxr-x 2 mininet mininet 4096 Oct  9 08:52 results
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Figure 11: Создание копии скрипта lab_iperf3_topo2.py и его дальнейшее помещение в подкаталог iperf

Добавление в скрипт настроек параметров производительности

Figure 12: Добавление в скрипт lab_iperf3.py записи об импорте time; снятие ограничений по использованию ресурсов процессора; добавление кода, чтобы каналы между хостами и коммутатором были по 100 Мбит/с с задержкой 75 мс, без потерь

Добавление в скрипт настроек параметров производительности

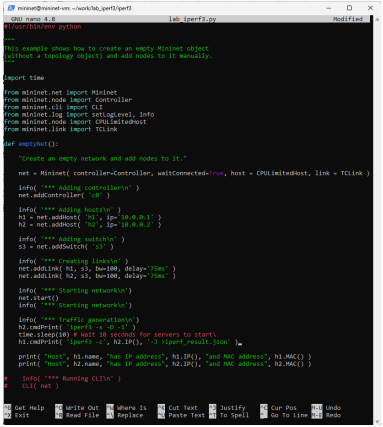

Figure 13: Описание запуска на хосте h2 сервера iPerf3, на хосте h1 запуска с задержкой в 10 секунд клиента iPerf3 с экспортом результатов в JSON-файл. Комментирование строк, отвечающих за запуск CLI-интерфейса

Добавление в скрипт настроек параметров производительности

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ***
Starting network
*** Configuring hosts
h1 (cfs -1/100000us) h2 (cfs -1/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ...(100.00Mbit 75ms delay) (100.00Mbit 75ms delay
)
*** Waiting for switches to connect
s3
*** Starting network
*** Traffic generation
*** h2 : ('iperf3 -s -D -1',)
*** h1 : ('iperf3 -c', '10.0.0.2', '-J >iperf_result.json')
Host h1 has IP address 10.0.0.1 and MAC address ce:17:60:7d:c2:5e
Host h2 has IP address 10.0.0.2 and MAC address 06:5b:9d:9b:1c:37
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Figure 14: Запуск скрипта lab_iperf3.py на отработку

Построение графиков по проводимому эксперименту

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ plot_iperf.sh iperf_result.json
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ touch Makefile
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ _
```

Figure 15: Построение графиков и создание Makefile для проведения всего эксперимента

mininet@mininet-vm: ~/work/lab_iperf3/iperf3

GNU nano 4.8 Makefile

```
all: iperf_result.json plot

iperf_result.json:
        sudo python lab_iperf3.py

plot: iperf_result.json
        plot_iperf.sh iperf_result.json

clean:
        -rm -f *.json *.csv
        -rm -rf results
```

Figure 16: Добавление скрипта в Makefile

Построение графиков по проводимому эксперименту

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ nano Makefile
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make clean
rm -f *.json *.csv
rm -rf results
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make clean
rm -f *.json *.csv
rm -rf results
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make
sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ***
Starting network
*** Configuring hosts
h1 (cfs -1/100000us) h2 (cfs -1/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ...(100.00Mbit 75ms delay) (100.00Mbit 75ms delay
)
*** Waiting for switches to connect
s3
*** Starting network
*** Traffic generation
*** h2 : ('iperf3 -s -D -1',)
*** h1 : ('iperf3 -c', '10.0.0.2', '-J >iperf_result.json')
Host h1 has IP address 10.0.0.1 and MAC address 5a:56:d8:93:ec:01
Host h2 has IP address 10.0.0.2 and MAC address fa:cb:92:66:1a:f0
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
.
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
plot_iperf.sh iperf_result.json
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Figure 17: Проверка корректности отработки Makefile

Вывод

- В ходе выполнения лабораторной работы познакомились с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получили навыки проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet